Multibranch Fusion: A Multibranch Attention Framework by Combining Graph Convolutional Network and CNN for Hyperspectral Image Classification

Xun Liu, Alex Hay-Man Ng, *Senior Member, IEEE*, Linlin Ge, *Member, IEEE*, Fangyuan Lei, *Member, IEEE*, and Xuejiao Liao

***Abstract*— Graph convolutional network (GCN) has attracted increasing attention in hyperspectral image (HSI) classification due to its capability to capture the long-range correlations between adjacent land covers. Most GCN-based HSI classification methods have been proposed to address the four limitations (shape-fixed kernel, massive calculations and parameters, limited classification ability with limited labeled samples, and difficulty to capture the long-term relationships of land covers) of convolutional neural networks (CNNs) by operating on superpixel-based nodes. However, the pixels in each superpixel share the spectral–spatial features, overlooking the unique characteristics of individual pixels. To address these limitations of GCN and CNN and fully exploit their advantages, we propose a novel multibranch attention framework (MFAF), in which the specially designed GCN and CNN branches learn the complementary spectral–spatial features. Specifically, we develop a multiscale attentional GCN to enhance the ability to understand the long-range correlations between land covers, accomplished by constructing the multiscale attentional adjacency matrix. Then, based on the two designs of the dual-branch depthwise separable convolution (DSC) and the attention-based residual block, we present a new complementary dual convolutional attention network that extracts more discriminative spectral–spatial features of pixels. Finally, we introduce an attention-based fusion pooling (AFP) mechanism to combine the features generated by different network branches. Extensive experimental evaluations on four public HSI datasets demonstrate that the proposed MFAF achieves better performance than several state-of-the-art methods, delivering superior and consistent results in terms of overall accuracy (OA), average accuracy (AA), and kappa coefficient (KAPPA).**

***Index Terms*— Attention mechanism, convolutional neural network (CNN), graph convolutional network (GCN), hyperspectral image (HSI) classification, residual connection.**

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Xun Liu is with the School of Information Engineering, Guangdong University of Technology, Guangzhou 510006, China (e-mail: liuxun.stf@gmail.com).

Alex Hay-Man Ng is with the School of Civil and Transportation Engineering and the School of Information Engineering, Guangdong University of Technology, Guangzhou 510006, China (e-mail: hayman.ng@gmail.com).

Linlin Ge is with the Geoscience and Earth Observing System Group (GEOS), School of Civil and Environmental Engineering, University of New South Wales (UNSW), Sydney, NSW 2052, Australia (e-mail: l.ge@unsw.edu.au).

Fangyuan Lei and Xuejiao Liao are with Guangdong Provincial Key Laboratory of Intellectual Property Big Data, Guangdong Polytechnic Normal University, Guangzhou 510665, China (e-mail: leify@gpnu.edu.cn; liaoxuej@163.com).

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I. Introduction

HYPERSPECTRAL image (HSI) contains hundreds of bands, captured across a continuous wavelength range. In contrast to multispectral images, HSI provides richer spectral–spatial information. This allows the identification and differential of land cover types with greater precision and accuracy. Owing to these advantages, HSI has been widely used in various applications such as environment monitoring, geological exploration, object tracking, and tree species classification [1], [2], [3], [4], [5], [6], [7], [8]. HSI classification is a critical premise for these applications [9], [10], [11], [13].

Over the past several decades, a variety of methods have been proposed for HSI classification. In the early stage, many traditional machine learning models, such as K-nearest neighbor classifier [14], linear regression (LR) [15], support vector machine (SVM) [16], and sparse representation [17], [18], were employed to extract the spectral and spatial information from HSI. However, these methods usually require handcrafted features and professional expertise, making it challenging to learn the robust deep feature representations of HSI.

Motivated by the limitations and the remarkable success of deep learning (DL) in a wide range of applications, researchers have attempted to apply DL models to HSI classification. Currently, owing to the inherent capability of DL to autonomously extract adaptive and robust deep features from HSI, DL methods, such as stacked autoencoders (SAEs) [19], recurrent neural networks (RNNs) [9], [20], and convolutional neural networks (CNNs) [21], [22], [23], [24], [25], have obtained promising results for HSI classification. SAEs [19] are the first successful attempt at introducing the concept of DL into HSI classification tasks. Subsequently, Hang et al. [20] propose RNN to effectively learn the relationship between adjacent spectral bands. Hu et al. [26] design a simple 1-D CNN with five layers to extract the spectral features. To extract the spatial and spectral features, Zhao and Du [27] propose a spectral–spatial feature-based classification (SSFC)

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framework by applying dimension reduction and 2-D CNN techniques. Furthermore, 3-D CNN [28] and multichannel CNN (MCNN) [29] have been proposed for learning hierarchical high-level features. To address the degradation [30] in deeper networks, SSRN [31] and FDSSC [32] introduce residual connection [30] and dense connection [33] to HSI, respectively. In addition, several attention networks such as DBDA (HSI with attention network) have also been applied to HSI classification [34]. However, the existing CNN-based models suffer from several limitations, including the following: 1) the shape-fixed kernel may blur the boundaries between adjacent land covers; 2) the models require massive number of calculations and parameters; 3) their classification ability with HSI is limited due to the small number of labeled samples; and 4) they struggle to directly capture the relationships of land covers at long-term distances.

Recently, inspired by the excellent performance of Vision Transformer (ViT) [44] in computer vision applications, Transformer has been applied in HSI classification [10], [12], [35], [45], [46]. He et al. [45] first propose a Transformer-based framework, named HSI-BERT, to capture the global dependence among pixels. To learn the locally spectral features from multiple neighboring bands, SpectralFormer [35] is proposed. Subsequently, Sun et al. [10] present a spectral–spatial feature tokenization transformer (SSFTT) to extract the high-level semantic features via a Gaussian-weighted feature tokenizer. The abovementioned models utilize Transformer instead of the convolution with shape-fixed kernel to model long-term image dependencies, which addresses the first and fourth limitations of CNN-based models. However, due to the abundant spectral bands of HSI, Transformer may ignore the influence of local neighboring bands to object classification [47]. In addition, the encoding of local fine-level spatial–spectral features is inefficient [12].

In addition to these Transformer-based models, graph convolutional networks (GCNs) [36] and their variant models garnered attention and have been employed for HSI classification. By encoding HSI into a graph, the long-range correlations between land covers can be effectively captured. Qin et al. [37] propose a spectral–spatial GCN (S2GCN) by learning the information of adjacency nodes and neighboring pixels. S2GCNs treat each pixel in HSI as a graph node, leading to a considerable computation load. Sha et al. [38] introduce an attention mechanism, based on the spatial–spectral similarity of pixels, into GCN to assign appropriate weights to different neighboring nodes. Bai et al. [39] design a deep attention GCN (DAGCN) by using the deep CNNs concepts of dense connections [33] and dilated convolutions [40] to extract the non-Euclidean features.

The aforementioned models are GCN-based HSI classification models. GCN has the following characteristics and advantages: 1) GCN is designed for semisupervised node classification; 2) GCN can deal with non-Euclidean structured data and effectively model the long-range correlations between nodes; 3) The parameters of GCN are significantly reduced compared to CNN, allowing GCN-based models to overcome the limitations of CNN models, specifically limitations 3 and 4, while also partially addressing limitation 2. Namely, they can capture the complex topological information of land covers with limited labeled HSI. However, encoding a large number of pixels contained in HSI as graph nodes can blur the boundaries between adjacent land covers and bring computational challenge. To address the issues, various superpixel-based GCNs [41], [42], [43], [48], [49], [50] are proposed. For example, MDGCN [41] reduces the number of nodes by using superpixels instead of individual pixels as graph nodes. Subsequently, Zhang et al. [50] construct a superpixel graph and develop multiadaptive receptive field-based graph neural framework to capture the spatial–spectral features. Considering that the size and shape of superpixels can be flexibly adjusted, these superpixel-based models can adaptively learn kernel parameters, addressing the first limitation of CNN-based models. However, a caveat remains: all pixels within a given superpixel share features, overlooking the unique characteristics of individual pixels.

To tackle the issues of both single CNN-based and GCN-based models, several fusion models combining CNN and GCN approaches have been proposed, in which the CNN extracts the features at the pixel level to avoid the neglect of individual pixels from single GCN-based models, while the GCN learns the long-range correlations and superpixel-level features between land covers to address the limitations of single CNN-based models. Liu et al. [61] propose a CNN- and GCN-based fusion framework (CEGCN) to extract the pixel-level and superpixel-level spatial–spectral features. To better extract these features, Zhou et al. [62] propose an attention multihop graph and multiscale convolutional fusion network called AMGCFN, which integrates a novel multiscale CNN and multihop GCN. These fusion models typically utilize several ordinary convolutional layers and multiple graph convolutional layers to capture the pixel-level and superpixel-level features, resulting in many parameters. This increases the risk of overfitting, especially with small training sample sizes.

To avoid overfitting while retaining the advantages of fusion models, we propose multibranch attention framework (MFAF), a novel attention framework that integrates a designed GCN and CNN into a single network (Fig. 1). Specifically, we first construct a superpixel-level graph to reduce the computational cost and facilitate the learning of adaptive kernel parameters. Second, we develop the spectral interaction network to eliminate noise and capture the spectral features of different bands. Then, a multiscale attentional GCN is designed to improve the ability of learning the long-range correlations between land covers, which addresses the limitations of CNN-based models. In the multiscale attentional GCN, the graph encoding and decoding are used to match the data between different structures. Next, we propose the complementary dual convolutional attention network to extract the local spectral–spatial features of pixels within HSI, which avoids the overlook of individual pixels in superpixel-based GCN models. Finally, the attention-based fusion pooling (AFP) is employed to fuse the features extracted by different network branches, aiming to enhance the performance of HSI classification.

In summary, the main contributions of this article are summarized as follows.

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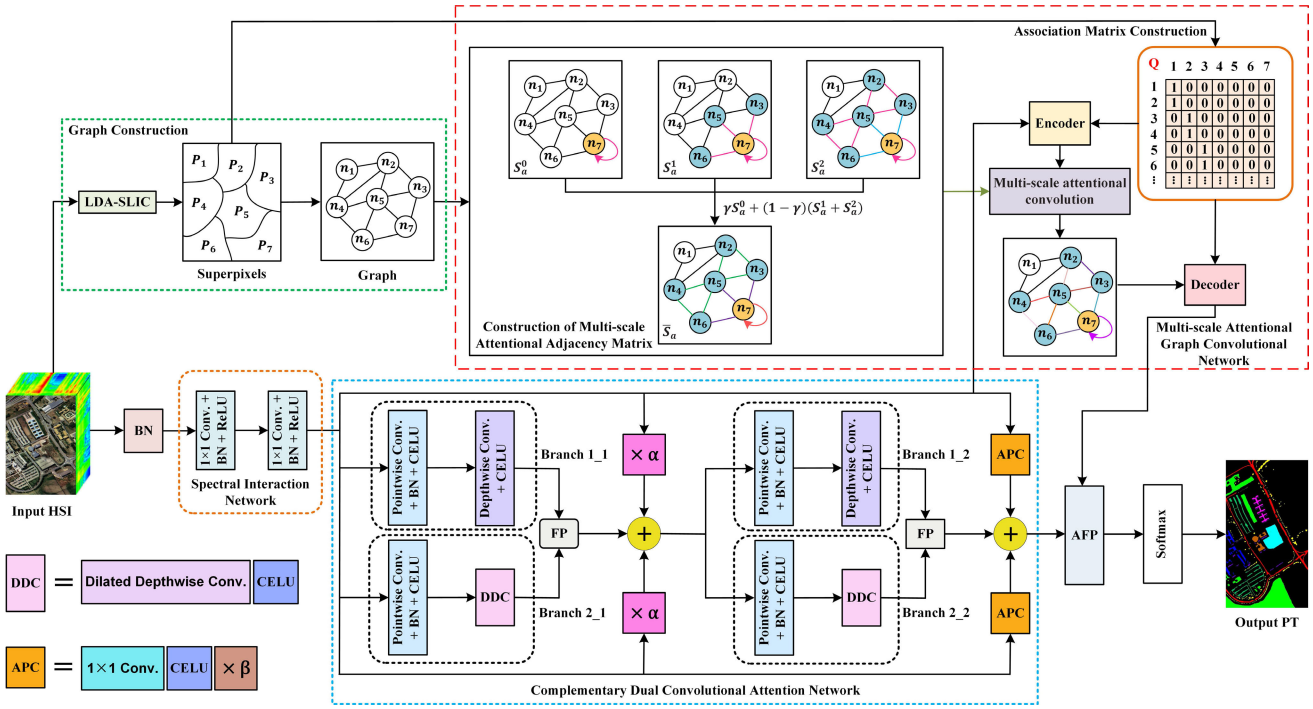


Fig. 1. Architecture of the proposed MFAF, which consists of three main components: a spectral interaction network to suppress the noise and reduce the band dimension, a multiscale attentional GCN to learn the inherent correlation between land covers, and a complementary dual convolutional attention network to extract the local spectral–spatial features.

1) *Multiscale Attentional GCN:* Addressing the limitations of CNN-based models, we design a novel multiscale attentional GCN. The network employs the proposed multiscale attentional convolution to more effectively extract superpixel-level features and learn the long-range correlations between land covers, compared to a classical graph convolution.
2) *Complementary Dual Convolutional Attention Network:* To address the oversight of individual pixels in superpixel-based GCNs, we introduce a new complementary dual convolutional attention network that utilizes two key designs to enhance the capability of capturing local pixel features in a complementary and dual branch manner: (D1) a dual-branch depthwise separable convolution (DSC) and (D2) an attention-based residual block.
3) *Integrated Framework:* To leverage the advantages and mitigate the limitations of different networks, we develop an attention-based fusion operation to better integrate the proposed CNN and GCN into a single framework, as compared to common fusion schemes such as column concatenation. This framework jointly learns robust superpixel and pixel-level discriminative features with limited labeled HSI.

The rest of this article is organized as follows. In Section II, we introduce relevant concepts, definitions, and GCNs. We propose our MFAF in Section III. Section IV evaluates the performance of MFAF. Section V summarizes this article.

II. Related Concepts and Definitions

In this section, we follow Kipf and Welling [36] to introduce some notations and definitions related to the proposed method. In addition, several GCNs that serve as the basis for the proposed model are described.

A. Notations and Descriptions

We use the node set $\mathcal{V}$, edge set $\mathcal{E}$, adjacency matrix A, and node feature matrix X of an undirected graph to denote the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, A, X)$. Each entry a_{km} in A denotes the edge weight between node v_k and node v_m. If the two nodes are directly connected, then $a_{\text{km}} = 1$, otherwise $a_{\text{km}} = 0$. The graph Laplacian matrix L is denoted as $L = D - A$, where D is the degree matrix of A. Its normalized version $L_{\text{sym}} = D^{-(1/2)} L D^{-(1/2)}$ is a symmetric positive semidefinite matrix. We can define the normalized and self-loops adjacency matrix S_{sym} as $S_{\text{sym}} = \tilde{D}^{-(1/2)} \tilde{A} \tilde{D}^{-(1/2)}$ by introducing the adjacency matrix with self-loops $\tilde{A}$, where $\tilde{D}$ is the self-loops version of D, and $\tilde{A} = A + I$ describing the edge relationships of considering self-loops. Here I is an identity matrix.

B. Graph Convolutional Network

Given a filter g_θ and a graph signal s, Bruna et al. [51] define the first-generation graph convolution using the graph Fourier transform and the spectral decomposition of Laplace matrix

$$g_\theta * s = U g_\theta(\Lambda) U^{\mathrm{T}} s \tag{1}$$

where U is the eigenvector matrix of the normalized Laplacian matrix, $\Lambda = \text{diag}[\lambda_1, \lambda_2, \ldots, \lambda_n]$ is the diagonal matrix of the eigenvalues $(\lambda_1, \lambda_2, \ldots, \lambda_n)$, and $g_\theta(\Lambda) = (\theta_1, \theta_2, \ldots, \theta_n)$ is a learnable diagonal matrix about the parameters $(\theta_1, \theta_2, \ldots, \theta_n)$. However, the convolution requires a massive number of calculations and parameters, and loses the

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spatial locality of the convolution kernel. To tackle these limitations, Defferrard et al. [52] redefine the graph convolution by cleverly designing $g_\theta(\Lambda) = \sum_{m=0}^{K} \alpha_m \Lambda^m$

$$g_\theta * s = U \sum_{m=0}^{K} \alpha_m \Lambda^m U^{\mathrm{T}} s = \sum_{m=0}^{K} \alpha_m {L_{\mathrm{sym}}}^m s \tag{2}$$

where $\alpha_0, \alpha_1, \ldots, \alpha_K$ are trainable parameters learned through backpropagation. We usually refer to the convolution of (2) as the second-generation graph convolution. To further reduce the calculations, the convolution kernel $g_\theta(\Lambda)$ is designed as a truncated Chebyshev polynomial $\sum_{m=0}^{K-1} \theta_m \mathrm{T}_m(\tilde{\Lambda})$, where $\theta_1, \theta_2, \ldots, \theta_m$ denote the coefficients of Chebyshev polynomial, and $\tilde{\Lambda} = (2/\lambda_{\max})\Lambda - I$. Here $\lambda_{\max}$ is the largest eigenvalue of L_{sym}. Based on the Chebyshev polynomial, the convolution becomes

$$g_\theta * s = \sum_{m=0}^{K-1} \theta_m \mathrm{T}_m(\tilde{L}_{\mathrm{sym}}) s \tag{3}$$

where $\tilde{L}_{\mathrm{sym}} = (2/\lambda_{\max})L_{\mathrm{sym}} - I$. The convolution is called the third-generation graph convolution. Let $K = 2, \lambda_{\max} = 2$, and $\theta_0 = -\theta_1 = \theta$, then the convolution can be simplified as follows:

$$g_\theta * s = \theta_0 s + \theta_1 (L_{\mathrm{sym}} - I)s = (I + D^{-1/2} A D^{-1/2}) s\theta\,. \tag{4}$$

We renormalize $I + D^{-1/2} A D^{-1/2}$ as $\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$ to weaken the gradient disappearance and gradient explosion of the network. Then the convolution can be rewritten as follows:

$$g_\theta * s = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} s\theta\,. \tag{5}$$

We introduce an activation function σ to (5) and provide its matrix form as follows:

$$H^{(l+1)} \leftarrow \sigma(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)}) \tag{6}$$

where $H^{(l)}$ is the input node representation at layer l with $H^{(l)} = X$ if $l = 1$, and $W^{(l)}$ is the weight matrix at layer l. Equation (6) is the well-known fourth-generation graph convolution proposed by Kipf and Welling [36]. By applying the convolution twice and using a softmax classifier to predict labels, a classic two-layer GCN can be expressed as follows:

$$Y_{\mathrm{GCN}} = \mathrm{softmax}(S_{\mathrm{sym}} \mathrm{ReLU}(S_{\mathrm{sym}} \mathrm{X} \mathrm{W}^{(1)}) W^{(2)}) \tag{7}$$

where ReLU denotes a nonlinear activation function, with $\mathrm{ReLU}(\cdot) = \max(0, \cdot)$, and $W^{(1)}$ and $W^{(2)}$ denote the trainable weight matrices updated through backpropagation.

III. Proposed Method

Although CNN-based HSI classification models can effectively consider the individual differences of local pixels and learn the local spectral–spatial features, they have several limitations as outlined in Section I, such as ignoring the inherent correlation between land covers. Recent GCN-based models enhance their capability to capture the correlation between nodes and avoid other limitations of CNN-based models by constructing superpixel-based graph. However, these approaches tend to neglect the distinct differences of individual local pixels. These observations motivate us to leverage the strengths of each network and compensate for the respective weaknesses by combining CNN and GCN into a single framework. In this article, we propose the MFAF, which includes five modules: 1) graph construction that built the superpixel-based graph; 2) spectral interaction network that eliminates the noise and reduces the spectral dimension; 3) multiscale attentional GCN that learns the inherent correlation between land covers; 4) complementary dual convolutional attention network that extracts the local spectral–spatial features of HSI at the pixel level; and 5) feature fusion and classification that can fuse the features extracted by different networks and predict the label of each pixel. In the following, we will describe each module in detail.

A. Graph Construction

Due to the large number of pixels in HSIs, it is unrealistic for graph convolution to deal with the complex computation of a vast graph generated by directly considering these pixels as the nodes in a graph. To address this limitation, the GCN-based models usually use a dimensionality reduction algorithm and work on superpixel-based nodes. Similarly, we use linear discriminant analysis (LDA) to reduce the spectral dimension and pixels of the HSI. In addition, we apply the simple linear iterative clustering (SLIC) [53] to partition the reduced pixels into superpixels, which significantly reduces the computational cost. The pixels contained in the corresponding superpixel are homogeneous and have strong similarity in spatial and spectral features. By considering each superpixel as a node and establishing the connection relationships between nodes, the HSI is converted to an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ and $\mathcal{E}$ are the node set and edge set of the HSI, respectively. The node feature matrix V of $\mathcal{G}$ represents the average spectral signatures of the involved pixels in each superpixel (node). We define the adjacency matrix A of $\mathcal{G}$ describing the connection relationship between nodes in the following equation:

$$A_{i,j} = \begin{cases} 1, & \text{if } P_i \text{ and } P_j \text{ are adjacent (directly connected)} \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

where $P_i = \{x_j^i\}_{j=1}^{n_i}$ is the ith superpixel. Here x_j^i is the jth pixel of P_i, and n_i denotes the number of pixels contained in P_i.

B. Spectral Interaction Network

Considering that the original HSI may contain some noise and redundant band information, the direct extraction of robust features from the HSI is limited. To address the limitation, we design a new spectral interaction network to suppress the noise and reduce the band dimension before extracting the features. In the spectral interaction network, we first apply batch normalization (BN) to normalize the HSI, and then use two consecutive 1×1 convolutions with the bias b_i^l to conduct the spectral interaction of HSI, which effectively eliminates the noise and controls the spectral information. Let the normalized HSI act as an input, and $X_i^{l+1}(p_1)$ is the output feature map of

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the lth 1×1 convolutional layer in the ith spectral channel. Then we have

$$X_i^{l+1}(p_1) = \mathrm{ReLU}(\mathrm{BN}(W_i^l \cdot X_i^l(p_1) + b_i^l)) \tag{9}$$

where $p_1 = (x, y)$ is the spatial location of the pixel, $X_i^l(p_1)$ is the value of the lth 1×1 convolutional layer in the ith input spectral channel for the location p_1, W_i^l is the trainable weight of the ith kernel with a size of 1×1 in the lth convolutional layer, and the operator $\cdot$ is the inner product.

C. Multiscale Attentional GCN

The proposed multiscale attentional GCN consists of these four stages: (S1) association matrix, (S2) graph encoder, (S3) multiscale attentional convolution, and (S4) graph decoder. For the sake of clarity, we describe each stage in detail.

(S1) Association Matrix: As described in Section III-A, we use the superpixels generated by LDA and SLIC instead of individual pixels to reduce the computational cost. To construct a bridge between pixel and superpixel, we define an association matrix Q as follows:

$$Q_{i,j} = \begin{cases} 1, & \text{if } x_i \in P_j \\ 0, & \text{otherwise} \end{cases} \tag{10}$$

where x_i is the ith pixel of the pixels (contained in the superpixel P_j) flattened by the spatial dimension.

(S2) Graph Encoder: After generating the association matrix, we encode the HSI transformed by the spectral interaction network into graph nodes

$$V = \mathrm{Encode}(X_t; Q) = Q^{\mathrm{T}}\mathrm{Flatten}(X_t) \tag{11}$$

where X_t denotes the feature matrix (transformed by the spectral interaction network) of HSI, $\mathrm{Encode}(\cdot; Q)$ denotes the graph encoder, and $\mathrm{Flatten}(\cdot)$ is an operator of flattening X_t by the spatial dimension.

(S3) Multiscale Attentional Convolution: By repeatedly applying the convolution in (6), a m-layer convolution can be expressed as follows:

$$\hat{H}_m = S_{\mathrm{sym}}\sigma(\cdots S_{\mathrm{sym}}\sigma(S_{\mathrm{sym}}\mathrm{X}\mathrm{W}^{(1)})W^{(2)})\cdots)W^{(m)}. \tag{12}$$

We know that the activation function σ can be ignored according to Wu et al. [54]. Therefore, we can rewrite the m-layer convolution by removing all the σ

$$\hat{H}_m = S_{\mathrm{sym}} \cdots S_{\mathrm{sym}} S_{\mathrm{sym}}\mathrm{X}\mathrm{W}^{(1)}W^{(2)} \cdots W^{(m)}. \tag{13}$$

By letting $W = W^{(1)}W^{(2)} \ldots W^{(m)}$, the simplified form of convolution in (13) is described as follows:

$$\hat{H}_m = {S_{\mathrm{sym}}}^m \mathrm{X}\mathrm{W} \tag{14}$$

where S_{sym} is the mth power of S_{sym}. We refer to the simplified convolution as the m-layer convolution of SGC [54]. Although the receptive field can be enlarged by increasing the value of m, SGC suffers from the oversmoothing when m is not less than 3 [55]. Limited to the oversmoothing, the classification performance needs to be improved. To address the problem, we propose our multiscale attentional convolution. For the multiscale attentional convolution, we utilize adjacency matrices with different powers to aggregate the multiscale neighboring nodes of HSI, and design a simple attention mechanism to adaptively adjust the contributions of their adjacency matrices to avoid oversmoothing. The proposed convolution updates the node features of HSI in four steps, i.e., construction of multiscale attentional adjacency matrix, multiscale feature propagation, multiscale linear transformation, and layer normalization (LN).

1) Construction of Multiscale Attentional Adjacency Matrix: To aggregate the neighboring nodes at different distances, we construct the multiscale attentional adjacency matrix $\bar{S}_a$ by combining the attention mechanism and adjacency matrices with different powers, which is designed as follows:

$$\bar{S}_a = \gamma S_a^0 + (1-\gamma)(S_a^1 + S_a^2 + \cdots + S_a^K) \tag{15}$$

where $S_a^0 = I$ is an identity matrix used to additionally introduce node's own features, γ is a learnable attention parameter of adjusting the weight of node's own features, and S_a^K denotes the Kth power of S_a, with $S_a = \tilde{D}^{-(1/2)}\tilde{A}\tilde{D}^{-(1/2)}$. In the multiscale attentional adjacency matrix, the feature information from nodes that are K-hops away can be obtained via $S_a^0, S_a^1, \ldots, S_a^K$, and the contributions of node's own features and neighboring node features to performance improvement are controlled by γ and $1-\gamma$, respectively.

2) Multiscale Feature Propagation: For multiscale feature propagation, we average the node feature matrix V by using the features of neighboring nodes constructed by $\bar{S}_a$

$$\bar{H} = \bar{S}_a V. \tag{16}$$

This step encourages similar predictions among locally connected nodes because of the smoothness of (16).

3) Multiscale Linear Transformation: After the multiscale feature propagation, the multiscale linear transformation can be described as follows:

$$H = \bar{H}W \tag{17}$$

where W denotes a weight matrix to be learned.

4) Layer Normalization: As shown in Section III-A, we construct the graph using superpixels generated by LDA and SLIC. Given that the number of pixels within each superpixel may vary, we apply LN [56] to normalize the transformed features according to (17). Then we have

$$H = \mathrm{LN}(H). \tag{18}$$

By combining (15) and (18), we conclude that the proposed multiscale attentional convolution takes the following form:

$$H = \mathrm{LN}((\gamma S_a^0 + (1-\gamma)(S_a^1 + S_a^2 + \cdots + S_a^K))\mathrm{V}\mathrm{W}. \tag{19}$$

Our convolution utilizes adjacency matrices with various powers and the attention mechanism to extract K-hops node features, which learns the inherent correlation between land covers effectively. Compared with GCN's convolution that learns one-hops node features, our convolution can capture node features that are K-hops away.

(S4) Graph Decoder: Since each pixel of HSI corresponds to a category, we classify each pixel instead of superpixel-based graph nodes. Therefore, the graph nodes need

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to be restored to pixels. We define the graph decoder to project the features of graph nodes to pixel features by QH

$$H_p = \text{Decode}(H; Q) = \text{QH}. \tag{20}$$

In the graph decoder, the graph features (graph space) can be projected back to the spatial dimension (image space).

D. Complementary Dual Convolutional Attention Network

Although 3-D convolution network with many layers can be used to extract the local spectral–spatial features from HSI, it introduces a significant number of parameters and computations. These parameters can lead to the risk of overfitting when dealing with limited samples. Recently, a recent surge of research interest has concentrated on the DSC of MobileNets [57]. As shown in Fig. 2(a), DSC proposes to decompose standard convolution into depthwise convolution (DC) [57] and pointwise convolution (PC) [57] for substantial reduction in the number of parameters and computations. DC uses a single filter of size $k \times k$ for each input channel. Let c represent the number of input channels, the convolution is equivalent to a $k \times k$ group convolution [58] with c groups. PC is a 1×1 convolution of combining the feature maps produced by the DC layer. DSC uses the technologies of BN and ReLU after each decomposed convolution to accelerate convergence and bring nonlinearity, which leads to the information loss caused by ReLU on feature maps with fewer channels. To mitigate the information loss, Sandler et al. [59] propose MobileNetV2. The DSC of MobileNetV2, referred to as DSCV2, is shown in Fig. 2(b). In DSCV2, we first implement PC for channel interaction, then apply DC to extract features. Subsequently, PC is used without nonlinear activation to learn the feature maps across channels. Inspired by the designs, we propose a novel complementary dual convolutional attention network, as illustrated in Fig. 1. The network includes our dual branch DSC and attention-based residual block. We begin by analyzing these two designs before exploring the overall network structure.

1) Dual Branch DSC: We construct a novel dual branch DSC via the designs of the dual branch structure and dilated separable convolution [Fig. 2(c)], and apply it to HSI classification. As described in Fig. 2(c), the proposed convolution consists of two branches. In branch 1, we utilize PC for spectral features extraction, and then extract spatial features via DC. In branch 2, we apply PC and the designed dilated depthwise convolution component (DDC) made up of dilated depthwise convolution and the activation function CELU [60] to extract the spatial–spectral features. The proposed dilated depthwise convolution is a new DC of combining dilated convolution [40] and DC, which can be regarded as DC with dilation factor d. In the proposed dual branch DSC, we introduce a feature fusion operator, termed FP, to fuse the features extracted by the two branches. Detailed definitions related to this will be discussed in the context of the proposed complementary dual convolutional attention network.

2) Attention-Based Residual Block: As shown in Fig. 3(a), the residual block of ResNet [30] proposes to consider all the features of the front layer to enhance the features propagation

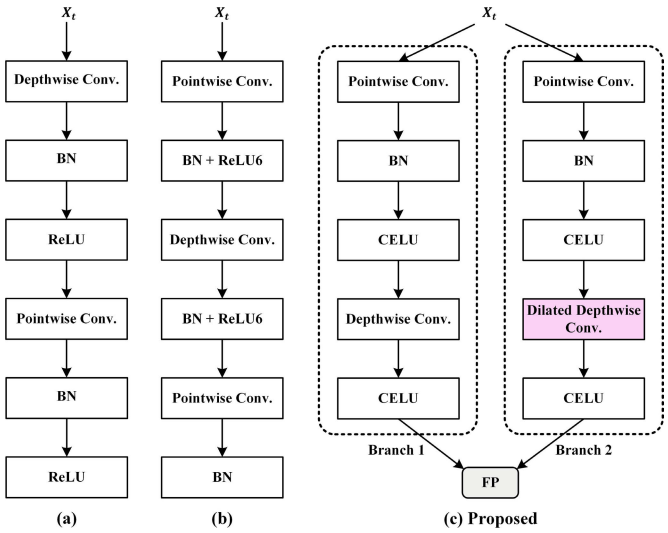


Fig. 2. Various DSCs. (a) DSC of MobileNets. (b) DSC of MobileNetV2. (c) Proposed dual branch DSC.

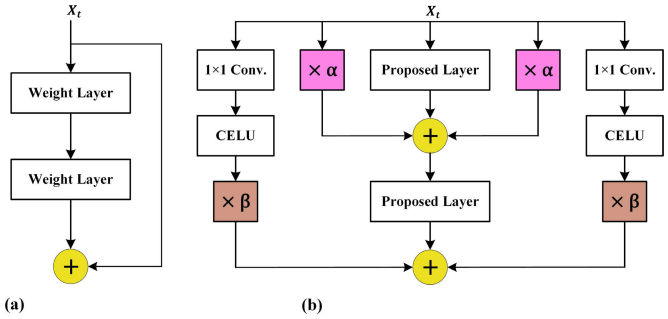


Fig. 3. Two residual blocks. (a) Residual block of ResNet. (b) Proposed attention-based residual block.

by introducing a direct edge. In order to enhance the feature reuse and feature propagation of HSI, we explore a new attention-based residual block [Fig. 3(b)]. From the figure, we observe that our attention-based residual block enhances the capability of learning HSI through two pairs of symmetrical skip connections with attention mechanism. We will introduce the details of our residual block in the context of the proposed network.

3) Structure of Complementary Dual Convolutional Attention Network: As illustrated in Fig. 1, the network mainly includes these parts of branch 1_1, branch 1_2, branch 2_1, branch 2_2, and the designed FP, APC, and attention factor α. For branch 1_1, we take the transformed feature matrix X_t as input, produce the output H_{1_1} after PC and DC, which is computed as follows:

$$H_{1_1} = \text{DC}(\text{PC}(X_t)). \tag{21}$$

For branch 2_1, we take X_t as input, then the output features H_{2_1} after PC and DDC can be described as follows:

$$H_{2_1} = \text{DDC}(\text{PC}(X_t)). \tag{22}$$

We design FP to fuse the features of H_{1_1} and H_{2_1}, which is defined as follows:

$$H_f = \text{FP}(H_{1_1}; H_{2_1}) = (H_{1_1} + H_{2_1})/2. \tag{23}$$

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After the feature fusion, we use the first pair of skip connections to enhance the feature propagation and feature reuse. Let H_d be the output features by the skip connections, then we have

$$H_d = H_f + \alpha X_t + \alpha X_t = H_f + 2\alpha X_t \tag{24}$$

where α is an attention factor. In contrast to the residual block of ResNet considering all the features of the front layer, we adaptively select the proportional features of the front layer via α. Similar to (21), in branch 1_2, the output features H_{1_2} after PC and DC can be expressed as follows:

$$H_{1_2} = \mathrm{DC}(\mathrm{PC}(H_d)) . \tag{25}$$

Analogous to (22), in branch 2_2, we write the output features H_{2_2} after PC and DDC as follows:

$$H_{2_2} = \mathrm{DDC}(\mathrm{PC}(H_d)) . \tag{26}$$

By using FP to combine the features of H_{1_2} and H_{2_2}, we have

$$H_a = \mathrm{FP}\left(H_{1_2}; H_{2_2}\right) = \left(H_{1_2} + H_{2_2}\right)/2 . \tag{27}$$

Due to the inconsistency in the number of feature channels between X_t and H_a, their features cannot be directly fused using the addition operator. To address the problem, we present APC, constructed using a 1×1 convolution, an activation function CELU, and an attention factor β as the second pair of skip connections. For each APC, we apply a 1×1 convolution to match the number of channels in X_t. Subsequently, the proportional features of X_t are determined by β. Let H_s be the output features of the proposed complementary dual convolutional attention network, then we have

$$H_s = H_a + \mathrm{APC}(X_t) + \mathrm{APC}(X_t) . \tag{28}$$

For the proposed complementary dual convolutional attention network, we notice that these parts complement each other, including branch 1_1 and branch 2_1, branch 1_2 and branch 2_2, and each pair of skip connections.

E. Feature Fusion and Classification

In the proposed multiscale attentional GCN, we learn that the inherent correlation between land covers at the superpixel level can be obtained using a limited set of labeled samples. However, such an approach might overlook the unique characteristics of individual pixels. In the complementary dual convolutional attention network, we aim to capture the local spectral–spatial features of HSI at the pixel level and address the drawback. By fusing the two different levels of features, the advantages of these two different networks can be well utilized and combined. Considering that the contributions of various networks to HSI classification may be different, an AFP is designed to fuse these features, which can be defined as follows:

$$H_{\mathrm{MFAF}} = \mathrm{AFP}(H_s; H_p) = (H_s + \mu H_p)/(1 + \mu) \tag{29}$$

where μ denotes an attention factor of adjusting the weights of H_p, which is learned by model training. CEGCN [61] and GPNet [63] introduce several fusion schemes used to combine features including column concatenation fusion, sum fusion, average fusion, and maximum fusion. Compared with these fusion schemes, the proposed AFP can adjust the contributions of the two different networks via μ. Let $\mu = 1$, then $\mathrm{AFP}(H_s; H_p) = (H_s + H_p)/2$. Therefore, the average fusion can be regarded as a special case of the proposed AFP.

To classify the label of each pixel, a softmax classifier is used to fuse the feature map H_{MFAF}, namely

$$Y = \frac{e^{P_i H_{\mathrm{MFAF}} + b_i}}{\sum_i^C e^{P_i H_{\mathrm{MFAF}} + b_i}} \tag{30}$$

where C is the number of land cover categories, and P_i and b_i denote the trainable parameter and bias. In this article, we use a cross-entropy error as the loss function, namely

$$\mathcal{L} = -\sum_{Z \in O_{\mathrm{label}}} \sum_{j=1}^{C} O_{zj} \ln Y_{zj} \tag{31}$$

where O is the label matrix, and Y_{zj} represents the probability of the zth pixel belonging to the jth category.

IV. Experiment

In this section, we conduct experiments to evaluate the effectiveness of the proposed MFAF. First, we compare MFAF with several deep learning-based methods on four HSI datasets in terms of per-class accuracy, overall accuracy (OA), average accuracy (AA), and Kappa coefficient (KAPPA). Then, we provide two analyses involving diverse training samples and various fusion schemes to illustrate the strengths of MFAF. Next, we compare the training and testing time of different methods to demonstrate the efficiency of MFAF. Finally, we conduct several ablation studies to validate the contributions of key designs.

A. Dataset

In this section, we introduce four widely used HSI datasets [50], i.e., Indian Pines, Pavia University, Salinas, and Houston 2013, to evaluate the performance of the proposed model.

1) Indian Pines: The first dataset, Indian Pines, was obtained by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor in north-western Indiana. It contains 220 spectral bands with a size of 145×145 and 16 land cover categories in the wavelength range from 0.4 to 2.5 μm. We retain 200 spectral bands by removing the noisy and water absorption bands of 104–108, 150–163, and 220. Moreover, 2.5%, 1%, and 96.5% of samples per class are randomly selected for training, validation, and testing, respectively. The details of the HSI are described in Table I.

2) Pavia University: The second dataset, Pavia University, was captured by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over the Pavia University. We use the reduced image with a size of 610×340, nine land cover categories, and 103 bands ranging from 430 to 860 nm. In addition, we randomly select 0.1%, 1%, and 98.9% of samples per class for training, validation, and testing, respectively. The details of the image are shown in Table II.

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TABLE I
DETAILED NUMBERS OF TRAINING, VALIDATION, AND TESTING PIXELS OF EACH CLASS ON INDIAN PINES DATASET

No.	Class name	Train.	Val.	Test.
1	Alfalfa	1	1	44
2	Corn-notill	36	14	1378
3	Corn-mintill	21	8	801
4	Corn	6	2	229
5	Grass-pasture	12	5	466
6	Grass-trees	18	7	705
7	Grass-pasture-mowed	1	0	27
8	Hay-windrowed	12	5	461
9	Oats	1	0	19
10	Soybean-notill	24	10	938
11	Soybean-mintill	61	24	2370
12	Soybean-clean	15	6	572
13	Wheat	5	2	198
14	Woods	32	13	1220
15	Buildings-Grass-Trees-Drives	10	4	372
16	Stone-Steel-Towers	2	1	90

TABLE II
DETAILED NUMBERS OF TRAINING, VALIDATION, AND TESTING PIXELS OF EACH CLASS ON PAVIA UNIVERSITY DATASET

No.	Class name	Train.	Val.	Test.
1	Asphalt	7	66	6558
2	Meadows	19	186	18444
3	Gravel	2	21	2076
4	Trees	3	31	3030
5	Painted metal sheets	1	13	1331
6	Bare Soil	5	50	4974
7	Bitumen	1	13	1316
8	Self-Blocking Bricks	4	37	3641
9	Shadows	1	10	936

3) Salinas: The third dataset, Salinas, was collected by the AVIRIS sensor in Salinas Valley, California. It contains 512 $\times$ 217 pixels of 16 crops. After removing the bands with serious water vapor absorption, 204 spectral bands ranging from 200 to 2400 nm are used. Then, 0.1%, 1%, and 98.9% of samples per class are randomly selected as the training, validation, and testing sets, respectively. The details of the dataset are illustrated in Table III.

4) Houston 2013: The fourth dataset, Houston 2013, was collected by the University of Houston campus. We use the image with a size of 349 $\times$ 1905, 15 land cover categories, and 144 bands ranging from 364 to 1046 nm. In addition, 0.25%, 1%, and 98.75% of samples per class are randomly selected as the training, validation, and testing sets, respectively. The details of the dataset are illustrated in Table IV.

B. Experimental Settings

We conduct all experiments using Adam [64] optimizer in PyTorch [65]. We now report the following best hyperparameters chosen in the proposed method. We set the learning rate lr according to various datasets, with $lr = 0.0001$ on Salinas, and $lr = 0.0005$ on other datasets. In the graph construction, the superpixel segmentation scale λ is fixed to 100 for each dataset. For the spectral interaction network, the number of filters per convolutional layer is 128. In the proposed multiscale attentional GCN, for all datasets, we set the number of neurons to 64, the maximum hop K to 2, and the initial attention parameter γ to 0.1. For the complementary

TABLE III
DETAILED NUMBERS OF TRAINING, VALIDATION, AND TESTING PIXELS OF EACH CLASS ON SALINAS DATASET

No.	Class name	Train.	Val.	Test.
1	Broccoli-green-weed1	2	20	1987
2	Broccoli-green-weed2	4	37	3685
3	Fallow	2	20	1954
4	Fallow-rough-plow	1	14	1379
5	Fallow-smooth	3	27	2648
6	Stubble	4	40	3915
7	Celery	4	36	3539
8	Grapes-untrained	11	113	11147
9	Soil-vineyard-develop	6	62	6135
10	Corn-Senesced-green-weeds	3	33	3242
11	Lettuce-romianes-4wk	1	11	1056
12	Lettuce-romianes-5wk	2	19	1906
13	Lettuce-romianes-6wk	1	9	906
14	Lettuce-romianes-7wk	1	11	1058
15	Vineyard-untrained	7	73	7188
16	Vineyard-vertical-trellis	2	18	1787

TABLE IV
DETAILED NUMBERS OF TRAINING, VALIDATION, AND TESTING PIXELS OF EACH CLASS ON HOUSTON 2013 DATASET

No.	Class name	Train.	Val.	Test.
1	Healthy-grass	3	13	1235
2	Stressed-grass	3	13	1238
3	Synthetic-grass	2	7	688
4	Tree	3	12	1229
5	Soil	3	12	1227
6	Water	1	3	321
7	Residential	3	13	1252
8	Commercial	3	12	1229
9	Road	3	13	1236
10	Highway	3	12	1212
11	Railway	3	12	1220
12	Parking-Lot-1	3	12	1218
13	Parking-Lot-2k	1	5	463
14	Tennis-Court	1	4	423
15	Running-Track	2	7	651

dual convolutional attention network, we set the number of filters to 128 in branch 1_1 and branch 2_1, and 64 filters in other parts. Besides, the kernel size of each DC is fixed to 7 $\times$ 7, the dilation factor d of each DDC with a kernel size of 3 $\times$ 3 is 2. We select the initial attention factors (α and β), with $\alpha = 0.25$ and $\beta = 0.4$ on all datasets. In AFP, we set μ to 0.7 on Indian Pines while setting $\mu = 0.9$ on other datasets. We train 120 epochs on Houston 2013 and 600 epochs on other datasets. Other hyperparameters refer to CEGCN [61]. All experiments of our method and baselines are repeated ten times. The experimental environment consists of an i9-7980XE CPU, Python 3.7, and a GTX-2080Ti GPU.

To verify the proposed model, we compare our MFAF with eleven state-of-the-art HSI classification methods including the deeper with contextual CNN (CDCNN) [64], the double-branch multiattention (DBMA) [65], the spectral–spatial residual network (SSRN) [31], the double-branch dual-attention network (DBDA) [34], the CNN-enhanced GCN (CEGCN) [61], the attention-based adaptive spectral–spatial kernel ResNet (A^2S^2K-ResNet) [66], the SSFTT [10], the fast dynamic GCN and CNN parallel network (FDGC) [67], the GCN and transformer fusion network (GTFN) [68], the attention multihop graph and multiscale convolutional fusion network (AMGCFN) [62],

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TABLE V

COMPARISON OF ALL METHODS IN TERMS OF OA, AA, AND KAPPA ON INDIAN PINES DATASET

Class	CDCNN	DBMA	SSRN	DBDA	CEGCN	A^2S^2K-ResNet	SSFTT	FDGC	GTFN	AMGCFN	AMS-M2ESL	MFAF
1	6.82±13.64	40.45±23.42	12.27±24.55	17.73±19.25	4.55±3.21	0.00±0.00	45.91±19.78	**82.67±15.03**	0.89±1.78	63.18±1.56	0.00±0.00	72.73±13.94
2	69.80±10.76	76.85±21.77	78.06±10.23	89.13±5.56	90.29±1.07	92.67±4.83	82.63±11.71	89.01±1.56	92.70±2.47	86.88±1.53	92.10±5.19	**93.57±0.44**
3	20.47±14.84	80.82±13.09	83.82±6.31	92.51±5.77	91.94±2.34	84.57±8.54	88.66±6.35	89.46±3.96	83.78±6.21	96.47±2.14	88.03±7.82	**98.48±0.75**
4	12.66±17.70	78.69±20.72	84.19±13.93	85.68±18.13	57.12±10.35	77.19±17.14	64.80±21.25	94.32±4.60	63.12±20.66	75.28±6.79	**96.98±3.34**	89.17±2.32
5	59.10±23.37	87.12±3.01	88.88±3.89	89.44±5.89	94.55±0.69	82.75±8.16	85.71±13.45	90.86±4.21	91.25±1.22	94.42±1.07	**99.65±0.44**	95.92±0.14
6	90.44±5.92	93.16±5.30	98.27±2.90	98.78±0.84	99.38±0.32	98.33±2.01	89.30±8.14	95.97±1.57	93.09±2.88	91.09±2.27	95.40±4.92	**99.01±0.53**
7	0.74±1.48	57.78±33.99	20.00±32.84	32.59±28.63	42.22±35.11	0.00±0.00	48.89±33.24	85.93±21.13	0.00±0.00	68.89±8.64	0.00±0.00	**87.41±3.78**
8	98.61±0.76	97.96±2.68	99.78±0.34	**100.0±0.00**	**100.0±0.00**	99.52±0.95	99.57±0.57	**100.0±0.00**	97.21±4.41	99.35±1.11	86.27±0.48	**100.0±0.00**
9	1.05±2.11	38.95±32.72	16.84±33.68	3.16±6.32	32.63±22.92	0.00±0.00	29.47±10.31	71.58±18.11	1.05±2.11	95.79±8.42	0.00±0.00	**97.89±2.58**
10	23.77±28.98	75.97±25.36	81.96±9.37	85.39±2.02	87.74±1.68	85.10±4.38	87.83±2.06	88.38±3.03	85.59±2.84	91.62±2.64	93.49±2.27	**95.29±1.51**
11	67.47±13.41	84.43±17.52	89.69±4.59	90.70±7.45	98.72±0.74	93.39±2.89	90.47±2.81	94.88±1.48	91.80±3.07	98.08±6.65	85.00±7.96	**99.18±0.24**
12	22.83±12.92	79.58±13.59	83.01±9.64	91.92±6.74	89.23±4.70	91.57±4.59	62.97±10.03	83.92±6.40	78.10±14.39	91.75±2.19	86.93±13.00	**97.59±0.43**
13	79.49±17.85	97.47±1.75	99.49±1.01	98.79±2.42	97.68±1.45	97.07±3.37	91.01±5.89	94.75±5.57	89.10±7.34	84.65±4.24	**99.90±0.20**	98.99±0.64
14	83.85±7.21	95.77±2.29	94.77±8.25	94.46±3.00	99.89±0.07	98.69±0.91	94.87±3.67	98.48±0.71	97.03±2.75	98.92±4.93	97.37±0.79	**99.95±0.07**
15	44.95±19.87	78.33±14.32	84.52±10.69	94.46±6.94	91.18±5.26	86.61±8.46	92.69±10.01	92.63±1.84	80.27±6.91	98.17±1.55	96.11±4.24	**99.84±0.32**
16	77.78±30.17	93.56±5.64	**97.33±4.80**	94.67±5.98	85.78±11.47	66.44±34.32	84.00±14.68	85.56±9.14	60.66±15.75	93.33±6.48	79.77±39.89	96.89±1.30
OA	59.62±4.90	84.40±8.52	87.41±3.20	91.18±2.23	93.37±0.63	90.70±1.77	87.06±2.65	92.63±0.42	88.59±1.47	93.83±0.84	90.35±2.55	**97.46±0.17**
AA	47.49±5.68	78.56±10.46	75.81±5.72	78.71±4.05	78.93±4.33	71.78±3.39	77.42±2.07	89.90±2.22	69.10±2.73	89.24±1.08	74.81±3.23	**95.12±1.05**
KAPPA	53.47±5.46	82.11±9.95	85.65±3.65	89.97±2.50	92.40±0.73	89.36±2.04	85.26±2.95	91.60±0.48	86.97±1.69	92.96±0.97	88.90±3.01	**97.10±0.19**

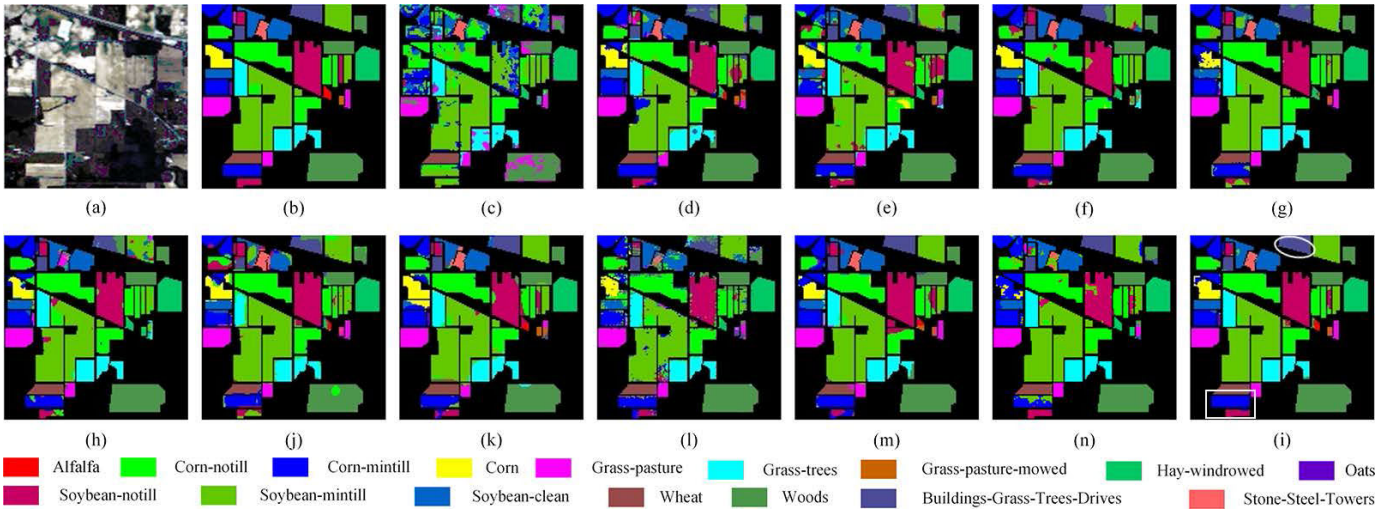


Fig. 4. False-color image, ground truth, and classification maps on Indian Pines dataset. (a) False-color image. (b) Ground truth. (c) CDCNN (OA = 59.62%). (d) DBMA (OA = 84.40%). (e) SSRN (OA = 87.41%). (f) DBDA (OA = 91.18%). (g) CEGCN (OA = 93.37%). (h) A^2S^2K-ResNet (OA = 90.70%). (j) SSFTT (OA = 87.06%). (k) FDGC (OA = 92.63%). (l) GTFN (OA = 88.59%). (m) AMGCFN (OA = 93.83%). (n) AMS-M2ESL (OA = 90.35%). (i) MFAF (OA = 97.46%).

TABLE VI

COMPARISON OF ALL METHODS IN TERMS OF OA, AA, AND KAPPA ON PAVIA UNIVERSITY DATASET

Class	CDCNN	DBMA	SSRN	DBDA	CEGCN	A^2S^2K-ResNet	SSFTT	FDGC	GTFN	AMGCFN	AMS-M2ESL	MFAF
1	88.84±7.62	92.72±3.12	89.48±6.83	93.99±4.41	89.82±2.57	87.33±5.25	84.57±4.38	67.44±8.84	83.50±4.08	76.11±5.93	84.02±2.90	**94.19±2.24**
2	95.29±2.44	95.88±1.92	95.51±2.23	95.41±3.84	98.73±1.37	92.76±4.12	98.51±1.87	96.70±2.64	93.95±6.17	99.27±0.50	92.44±2.36	**99.34±0.16**
3	3.59±4.97	37.09±17.77	40.52±11.25	35.49±9.38	53.43±19.21	51.83±9.53	58.57±13.31	66.89±8.73	**68.60±9.07**	41.16±10.56	55.18±13.86	64.88±4.24
4	43.79±29.97	65.14±18.19	71.75±12.45	86.38±5.41	65.39±6.02	63.53±18.06	61.76±1.80	61.08±11.66	62.44±15.86	67.49±5.67	**88.39±10.36**	76.72±3.21
5	57.08±28.55	96.86±2.67	99.44±0.73	95.31±8.89	99.80±0.27	97.45±3.59	98.65±1.40	96.71±4.39	96.68±3.39	97.01±2.93	99.92±0.00	**99.98±0.03**
6	25.81±10.99	57.23±21.87	73.66±11.72	72.96±16.71	94.73±3.31	48.68±18.60	76.58±5.82	92.60±4.74	80.05±7.73	**95.45±1.97**	93.34±4.16	89.92±1.49
7	4.33±8.66	42.10±30.14	38.59±35.21	29.36±18.10	69.85±22.43	5.82±3.79	63.19±8.51	79.29±12.78	47.51±8.55	88.71±5.91	**97.67±2.85**	92.20±6.48
8	85.55±7.27	78.73±18.58	89.65±12.08	**96.98±1.76**	94.63±1.30	48.93±8.53	52.06±4.53	72.07±13.27	59.69±9.94	84.04±5.67	57.21±5.66	96.22±1.24
9	72.29±37.92	93.59±3.79	**98.93±0.95**	95.75±4.97	56.50±6.94	32.80±27.20	27.35±7.97	18.29±8.43	74.06±12.77	47.26±12.87	95.48±3.46	97.01±1.57
OA	72.56±4.25	82.59±4.18	85.53±2.28	87.05±2.33	90.12±1.92	74.98±2.40	82.51±0.99	83.27±0.77	82.45±2.85	87.24±0.51	86.12±1.17	**93.60±0.47**
AA	52.95±9.90	73.26±7.46	77.50±4.14	77.96±4.55	80.32±4.16	58.79±3.51	69.03±1.33	72.34±3.23	74.05±1.74	77.39±1.74	84.85±1.51	**90.05±0.98**
KAPPA	61.59±6.42	76.23±6.14	80.51±3.12	82.60±3.25	86.77±2.56	66.01±3.03	76.36±1.19	77.89±1.12	76.45±3.49	82.88±0.73	81.40±1.61	**91.43±0.64**

and the adaptive mask sampling and manifold to Euclidean subspace learning (AMS-M2ESL) [69].

C. Comparison of Classification Performance

In this section, we evaluate the performance of MFAF quantitatively and qualitatively in comparison to the aforementioned competing models.

1) Results on Indian Pines Dataset: Table V reports the quantitative results of all methods in terms of OA, AA, and KAPPA on Indian Pines dataset. From the results, we make the following observations. First, the networks of DBMA, SSRN, DBDA, A^2S^2K-ResNet, SSFTT, FDGC, GTFN, and AMS-M2ESL that utilize attention mechanisms, residual connections, or multibranch learning significantly outperform CDCNN in terms of OA, AA, and KAPPA. This shows that the

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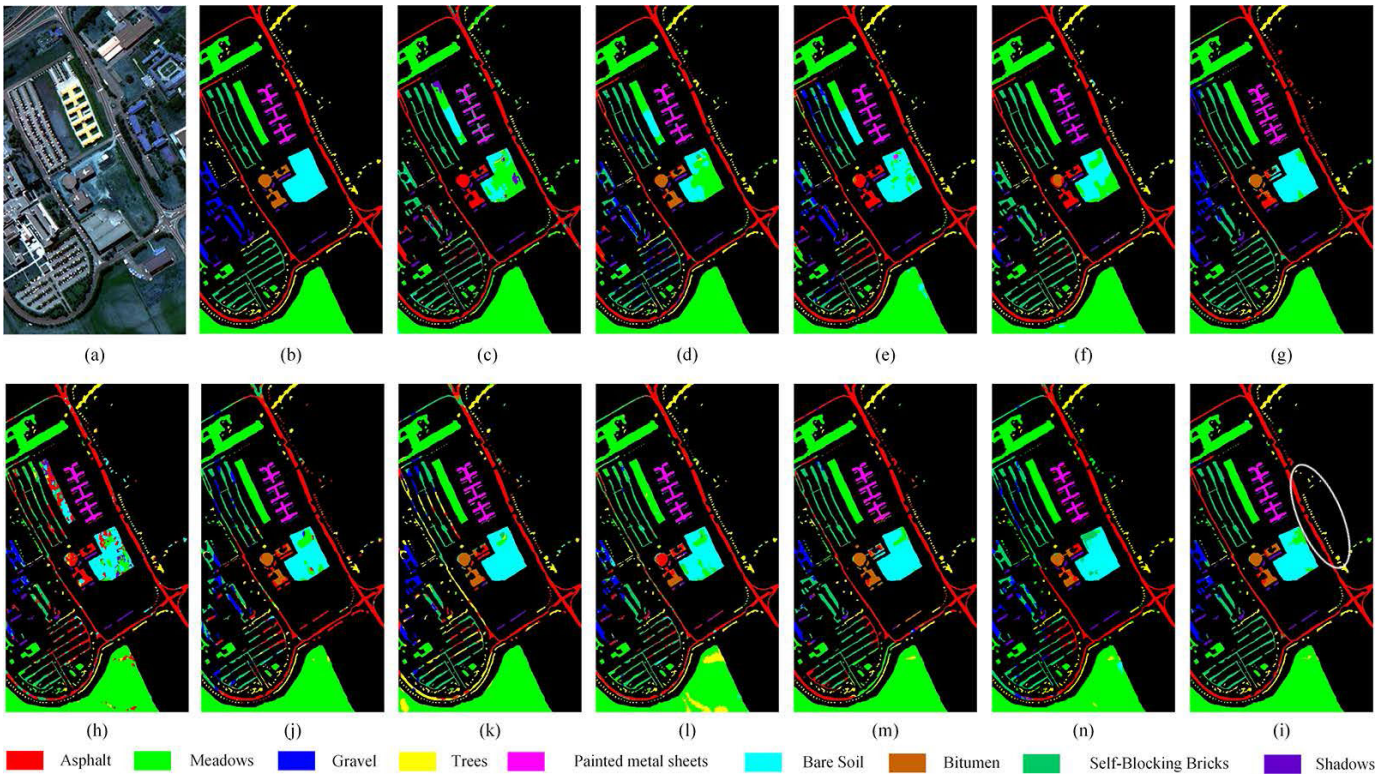


Fig. 5. False-color image, ground truth, and classification maps on Pavia University dataset. (a) False-color image. (b) Ground truth. (c) CDCNN (OA = 72.56%). (d) DBMA (OA = 82.59%). (e) SSRN (OA = 85.53%). (f) DBDA (OA = 87.05%). (g) CEGCN (OA = 90.12%). (h) A^{2}S^{2}K-ResNet (OA = 74.98%). (j) SSFTT (OA = 82.51%). (k) FDGC (OA = 83.27%). (l) GTFN (OA = 82.45%). (m) AMGCFN (OA = 87.24%). (n) AMS-M2ESL (OA = 86.12%). (i) MFAF (OA = 93.60%).

TABLE VII

COMPARISON OF ALL METHODS IN TERMS OF OA, AA, AND KAPPA ON SALINAS DATASET

Class	CDCNN	DBMA	SSRN	DBDA	CEGCN	A^2S^2K-ResNet	SSFTT	FDGC	GTFN	AMGCFN	AMS-M2ESL	MFAF
1	5.29±10.59	99.09±1.09	74.63±29.80	79.37±39.63	99.24±1.18	**100.0±0.00**	99.51±0.68	98.55±1.89	96.90±2.87	99.14±0.97	98.25±2.85	99.94±0.08
2	98.59±1.74	80.12±39.70	99.98±0.03	**100.0±0.00**	**100.0±0.00**	98.76±1.74	88.50±7.05	94.14±8.69	96.30±4.59	96.40±2.16	99.41±0.54	**100.0±0.00**
3	35.24±29.57	59.01±22.21	69.66±17.89	69.36±31.84	45.71±1.45	91.12±9.93	77.10±18.61	93.45±8.96	92.16±5.73	79.79±5.33	**100.0±0.00**	45.40±2.08
4	69.66±36.50	75.84±37.58	95.97±4.91	**97.87±1.75**	82.65±12.68	78.17±15.75	83.77±10.20	67.89±21.99	75.05±10.71	91.79±6.74	96.54±0.77	90.63±4.66
5	83.59±15.90	90.80±5.99	95.39±3.34	94.31±1.53	**98.23±1.22**	98.10±1.80	86.70±5.11	94.99±2.05	94.61±4.36	91.10±2.00	89.15±11.12	94.96±1.09
6	99.65±0.34	99.12±0.85	99.49±0.63	99.78±0.39	99.69±0.35	99.73±0.35	**100.0±0.00**	96.17±3.85	97.68±1.94	99.98±0.02	99.66±0.43	**100.0±0.00**
7	99.11±0.60	98.85±1.29	99.64±0.38	99.97±0.04	99.94±0.08	99.82±0.21	97.02±1.69	98.41±1.68	99.32±0.53	98.73±0.40	98.17±2.09	**99.98±0.02**
8	82.86±25.38	82.09±14.01	75.68±17.27	90.22±6.75	**93.76±3.41**	85.59±9.90	91.75±3.29	87.48±4.44	85.51±6.13	79.43±5.85	88.93±3.37	91.82±1.15
9	99.88±0.19	98.39±1.27	99.02±1.36	99.71±0.56	**100.0±0.00**	**100.0±0.01**	99.89±0.07	99.74±0.26	98.80±2.02	**100.0±0.00**	98.28±1.89	**100.0±0.00**
10	24.55±23.10	83.42±6.73	89.77±1.07	86.38±6.24	82.20±5.27	85.58±10.42	71.12±2.62	87.78±9.09	79.78±14.40	81.86±3.89	**97.99±1.35**	82.55±2.18
11	9.60±19.20	83.13±19.83	85.38±11.41	67.94±38.34	70.15±15.43	77.05±22.78	**99.00±0.87**	79.81±37.50	92.11±7.05	82.75±7.35	98.88±1.88	85.02±5.64
12	37.70±40.02	97.61±2.86	92.19±11.62	98.03±3.36	99.69±0.42	90.50±8.00	97.11±0.79	81.84±10.21	80.39±21.08	**99.87±0.25**	94.27±8.91	99.62±0.53
13	0.24±0.43	90.38±14.52	97.06±1.63	76.38±38.41	94.44±4.58	81.52±17.19	**97.90±1.24**	85.81±10.61	71.37±16.49	92.52±4.18	77.10±25.02	97.64±1.88
14	0.00±0.00	88.00±10.14	66.43±33.57	**98.88±0.45**	91.13±8.23	88.02±10.92	86.37±4.07	85.27±7.50	76.07±20.51	97.67±1.01	75.04±22.19	92.06±4.65
15	24.12±36.34	60.67±21.39	59.08±28.55	52.41±18.93	76.29±12.47	72.20±10.38	86.11±4.64	88.12±7.15	75.52±6.35	88.48±4.77	81.94±5.28	**92.22±2.90**
16	19.65±25.81	84.87±8.06	81.23±8.93	92.52±6.82	99.81±0.38	84.11±13.47	90.91±3.03	87.76±9.53	96.20±1.91	98.87±0.70	**100.0±0.00**	99.27±0.56
OA	63.66±5.31	84.36±3.23	84.36±1.33	87.10±6.08	90.99±1.11	89.42±1.63	91.01±0.74	90.89±1.47	88.62±1.88	90.76±1.29	92.58±1.95	**93.30±0.36**
AA	49.36±9.00	85.71±2.95	86.29±3.27	87.69±10.48	89.56±1.12	89.39±2.86	90.80±1.54	89.20±3.52	87.99±2.58	92.40±1.26	**93.35±3.22**	91.95±0.31
KAPPA	58.63±6.24	82.56±3.58	82.53±1.52	85.54±6.89	89.95±1.26	88.21±1.80	89.95±0.83	89.86±1.63	87.28±2.13	89.73±1.42	91.74±2.18	**92.39±0.41**

attention mechanisms, residual connections, and multibranch learning can help improve the performance of HSI classification. Second, these fusion networks (CEGCN and AMGCFN) with the superpixel-based GCN and CNN are superior to other baselines. This is because CEGCN and AMGCFN can better learn the features via the complementarity between GCN and CNN. Finally, by fusing the proposed multiscale attentional GCN and CNN with the dual branch DSC and attention-based residual block, our MFAF achieves the top-level performance and standard deviation among all competing methods in terms of OA, AA, and KAPPA. This infers that MFAF is more stable and effective than the compared methods.

Fig. 4 shows a visual comparison of classification maps obtained by the aforementioned methods on Indian Pines dataset. A visual inspection reveals that MFAF achieves fewer misclassifications compared with competitors. Meanwhile, in the classes of Alfalfa, Grass-pasture-mowed, and Oats, due to the lack of training samples, the pixels of these classes are prone to misclassification.

2) Results on Pavia University Dataset: Table VI lists the quantitative results of the proposed MFAF in comparison with other competing methods on Pavia University dataset. As can be inferred from the table, the proposed MFAF achieves SOTA performance and standard deviation, outperforming

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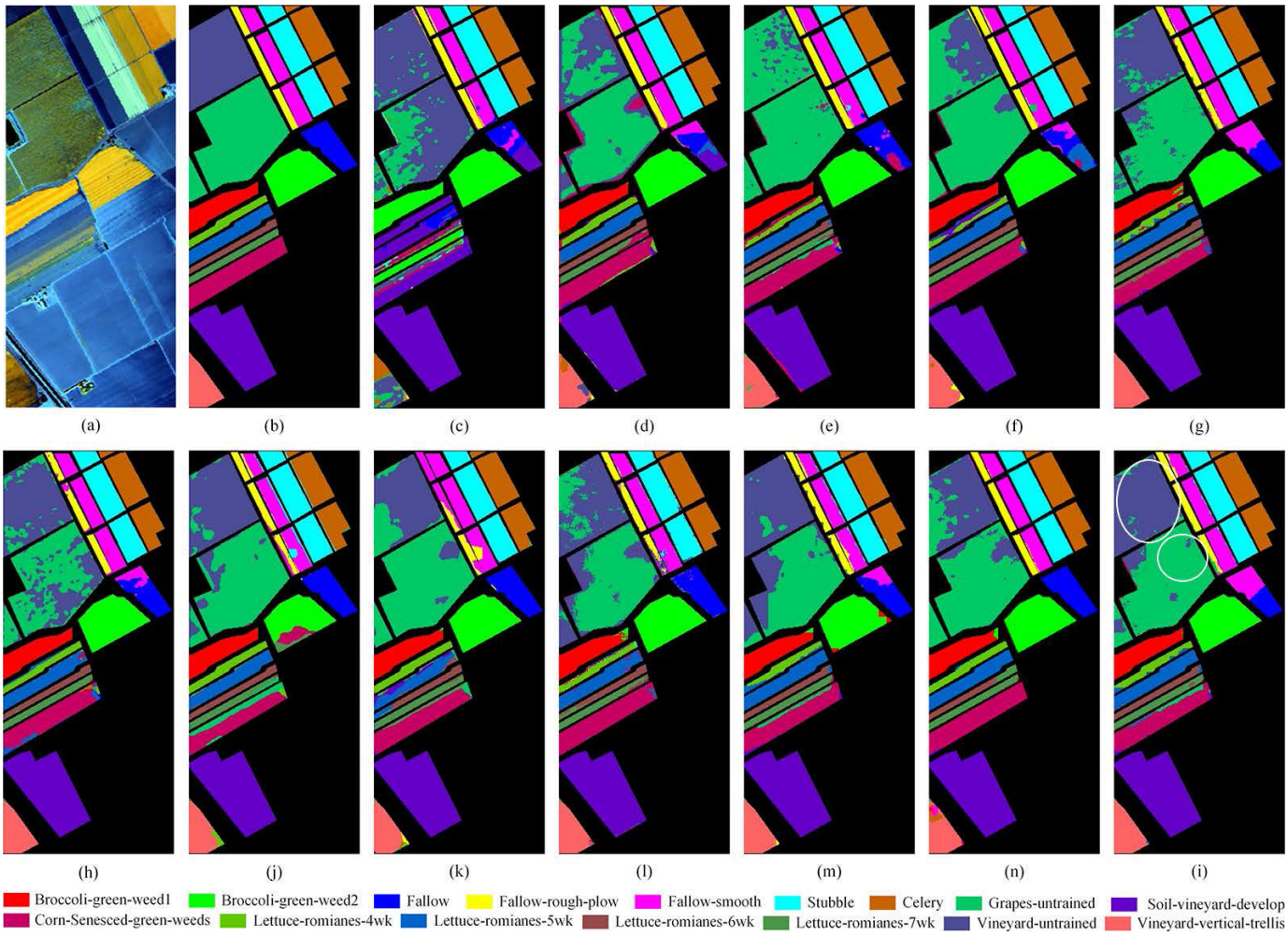


Fig. 6. False-color image, ground truth, and classification maps on Salinas dataset. (a) False-color image. (b) Ground truth. (c) CDCNN (OA = 63.66%). (d) DBMA (OA = 84.36%). (e) SSRN (OA = 84.36%). (f) DBDA (OA = 87.10%). (g) CEGCN (OA = 90.99%). (h) A^2S^2K-ResNet (OA = 89.42%). (j) SSFTT (OA = 91.01%). (k) FDGC (OA = 90.89%). (l) GTFN (OA = 88.62%). (m) AMGCFN (OA = 90.76%). (n) AMS-M2ESL (OA = 92.58%). (i) MFAF (OA = 93.30%).

all comparison methods in terms of OA, AA, and KAPPA. Specifically, MFAF improves over CEGCN by 3.86%, 12.11%, and 5.37%, improves over SSFTT by 13.44%, 30.45%, and 19.74%, and improves over AMGCFN by 7.29%, 16.36%, and 10.32% in terms of OA, AA, and KAPPA, respectively. These results demonstrate the strengths of the proposed MFAF. As shown in Fig. 5, MFAF is noticeably closer to the ground-truth map than the comparison methods, which further verify the effectiveness of MFAF with the complementary designs between the proposed GCN and CNN.

3) Results on Salinas Dataset: Table VII presents the quantitative results of different methods on Salinas dataset. The results show that the proposed MFAF performs better than CEGCN, SSFTT, FDGC, AMGCFN, and AMS-M2ESL, and outperforms other competitors by a substantial margin, which again demonstrates the advantage of MFAF. The visual inspection in Fig. 6 shows that MFAF achieves fewer misclassifications than the comparison methods. In addition, from the inspection, we observe that many pixels in the class of Grapes-untrained are misclassified into the class of Vineyard-untrained for different classification maps owing to the similar spectral signatures of the two land cover types.

4) Results on Houston 2013 Dataset: Table VIII shows the quantitative results of the proposed MFAF and comparison methods on Houston 2013 dataset. Similar to these results on Indian Pines, Pavia University, and Salinas datasets, the results show that the proposed MFAF outperforms all competitors by a substantial margin, which further supports the design of MFAF. The visual inspection in Fig. 7 shows that MFAF achieves more accurate classification map compared to all comparison methods.

D. Analysis of All Methods With Different Numbers of Training Samples

In this section, we investigate the OA performance of different methods under various percentages of training samples per class on Indian Pines, Pavia University, Salinas, and Houston 2013 datasets. For Indian Pines, the proportions of training samples per class are 1%, 2.5%, 5%, and 10%. For Houston 2013, the proportions of training samples per class are 0.125%, 0.25%, 0.5%, 1%, and 1.5%. On other datasets, 0.05%, 0.1%, 0.25%, 0.5%, and 1% samples per class are randomly selected as training samples. For all methods on all datasets, we set

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TABLE VIII
COMPARISON OF ALL METHODS IN TERMS OF OA, AA, AND KAPPA ON HOUSTON 2013 DATASET

Class	CDCNN	DBMA	SSRN	DBDA	CEGCN	A^2S^2K-ResNet	SSFTT	FDGC	GTFN	AMGCFN	AMS-M2ESL	MFAF
1	68.06±33.35	**89.26±8.38**	86.09±3.35	84.76±1.05	79.24±6.62	76.06±8.18	72.23±4.96	76.94±7.81	74.74±4.88	63.97±7.30	83.49±3.91	83.74±0.96
2	13.30±13.36	**86.11±6.02**	66.37±17.79	79.87±7.14	72.81±8.93	82.33±10.75	82.78±6.74	70.89±7.13	59.33±12.73	72.37±6.23	80.22±9.64	81.60±5.19
3	46.42±38.91	90.58±4.30	89.01±15.92	99.30±0.75	**99.97±0.06**	99.65±0.49	98.14±1.24	94.22±5.69	98.50±1.39	98.11±1.54	95.19±9.62	99.91±0.07
4	91.20±4.41	77.09±13.63	90.46±2.57	88.76±4.77	84.21±3.60	77.40±24.24	65.92±9.45	70.77±14.98	68.72±5.07	62.18±14.95	82.90±12.93	**94.11±0.26**
5	65.00±38.03	93.02±4.94	88.02±15.90	92.36±11.23	**100.0±0.00**	92.60±8.78	95.01±7.70	83.80±25.45	94.59±7.21	99.10±1.79	87.75±6.36	**100.0±0.00**
6	0.81±1.62	65.79±12.92	56.20±30.17	51.65±35.63	60.19±23.83	38.01±31.12	11.84±0.00	50.72±31.70	50.80±36.23	35.76±28.76	**100.0±0.00**	81.00±4.13
7	51.25±21.52	72.88±10.70	68.43±10.58	54.63±27.88	55.86±8.34	44.38±21.49	59.76±7.88	52.14±14.10	49.87±11.11	36.84±2.54	55.80±17.57	**73.99±5.74**
8	14.03±14.46	43.87±8.28	40.52±6.92	37.93±10.96	36.52±7.20	43.45±13.09	27.26±5.95	30.30±11.69	30.65±11.44	42.72±4.60	**84.82±5.59**	39.12±4.72
9	26.13±27.73	**73.01±10.17**	27.48±20.24	70.47±12.11	69.21±5.73	44.22±4.63	39.92±3.57	39.22±13.18	57.52±16.22	41.70±9.97	45.14±9.66	67.86±6.30
10	10.73±12.97	36.11±4.87	32.41±9.58	49.83±14.85	73.35±0.00	50.54±17.24	51.67±4.95	51.57±14.40	52.61±8.87	**79.74±9.45**	60.16±14.55	73.35±0.00
11	12.97±25.93	59.82±17.42	53.39±13.36	55.46±18.08	**78.52±3.07**	76.85±9.65	70.97±11.62	53.08±7.62	62.87±15.91	75.36±8.16	51.07±9.50	75.25±2.08
12	41.92±7.84	39.21±14.77	29.84±16.75	**62.04±6.63**	49.21±0.66	61.71±9.27	46.08±1.85	48.57±5.22	41.43±3.21	49.72±4.29	45.07±8.54	48.78±0.57
13	5.53±6.82	45.05±31.89	30.32±29.87	21.21±19.31	0.91±0.42	13.65±20.70	61.34±14.92	76.16±11.50	57.82±10.55	77.11±6.75	**81.00±9.53**	39.22±8.32
14	0.00±0.00	71.39±29.77	56.93±32.70	99.39±0.84	91.73±8.48	86.52±12.17	**100.0±0.00**	97.87±1.61	98.22±1.18	90.17±2.98	95.88±5.97	99.86±0.28
15	36.34±34.12	94.81±8.94	93.98±9.65	98.10±2.29	98.86±0.95	99.97±0.06	**100.0±0.00**	85.35±14.30	97.14±2.95	92.60±6.29	86.16±10.76	99.97±0.06
OA	36.68±3.93	68.85±4.62	60.40±5.77	69.55±3.40	70.80±1.09	66.51±3.62	64.64±0.96	62.21±2.02	63.61±3.26	65.96±3.15	68.09±3.57	**75.98±0.23**
AA	32.25±3.16	69.20±6.48	60.63±8.85	69.72±3.32	70.04±1.36	65.82±3.61	65.53±0.88	65.44±1.69	66.32±3.04	67.83±3.42	75.64±2.77	**77.18±0.31**
KAPPA	31.33±4.13	66.32±5.00	57.11±6.36	67.04±3.68	68.39±1.18	63.74±3.93	61.73±1.04	59.19±2.16	60.67±3.50	63.27±3.34	65.47±3.86	**74.04±0.27**

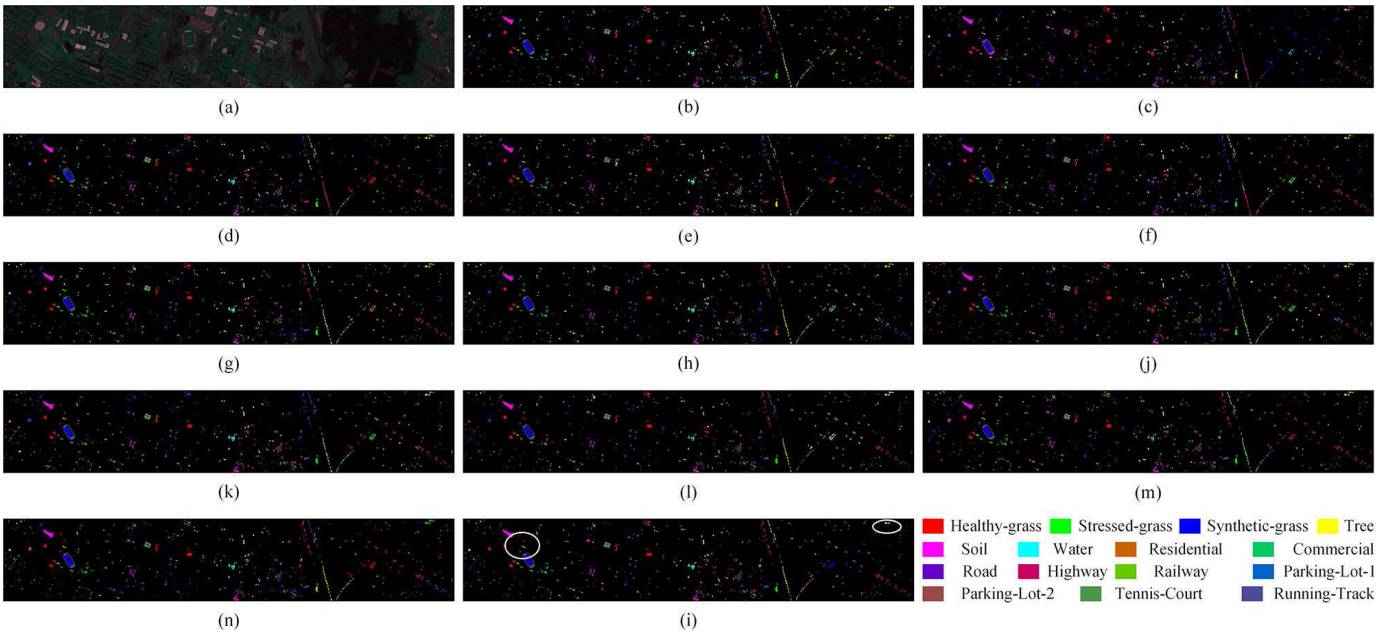


Fig. 7. False-color image, ground truth, and classification maps on Houston 2013 dataset. (a) False-color image. (b) Ground truth. (c) CDCNN (OA = 36.68%). (d) DBMA (OA = 68.85%). (e) SSRN (OA = 60.40%). (f) DBDA (OA = 69.55%). (g) CEGCN (OA = 70.80%). (h) A^2S^2K-ResNet (OA = 66.51%). (j) SSFTT (OA = 64.64%). (k) FDGC (OA = 62.21%). (l) GTFN (OA = 63.61%). (m) AMGCFN (OA = 65.96%). (n) AMS-M2ESL (OA = 68.09%). (i) MFAF (OA = 75.98%).

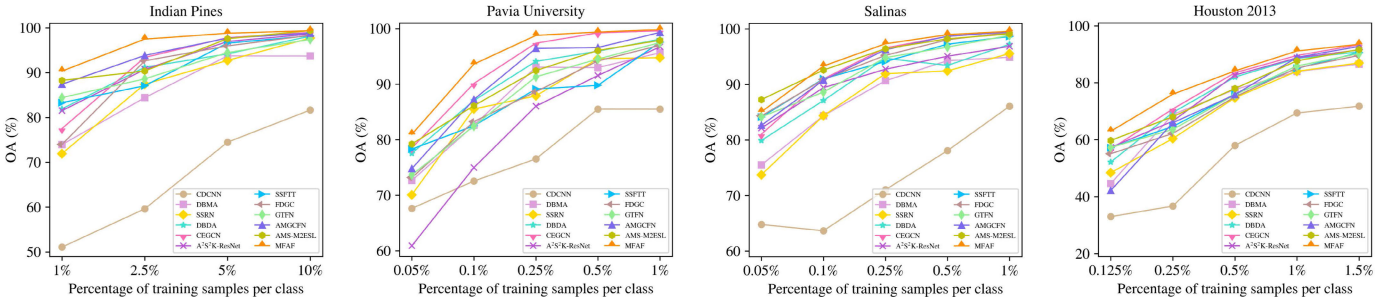


Fig. 8. Results of different methods under various percentages of training samples per class on Indian Pines, Pavia University, Salinas, and Houston 2013 datasets.

the percentage of validation samples to 1%. The OA results obtained by the 11 methods are reported in Fig. 8. It is observed that the OA results of the 11 methods improve as the percentage of training samples increases in most cases. We find that in a few cases, the OA results with more training samples reduce among several comparative methods. These abnormal results are mainly due to the extra noise brought by the increase in training samples. Meanwhile, the OA results of CEGCN, AMGCFN, and the proposed MFAF improve with the increase in the number of training samples, which

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TABLE IX

RUNNING TIME OF DIFFERENT METHODS ON INDIAN PINES, PAVIA UNIVERSITY, SALINAS, AND HOUSTON 2013 DATASETS. MS DENOTES MILLISECOND. $MFAF_B$ IS THE MODIFIED METHOD OF MFAF THAT EXCLUDES THE MULTISCALE ATTENTIONAL GCN, WHICH ACHIEVES THE OA OF 95.57%, 87.35%, 86.80%, AND 74.64% ON INDIAN PINES, PAVIA UNIVERSITY, SALINAS, AND HOUSTON 2013 DATASETS, RESPECTIVELY. OOM: OUT OF MEMORY

Dataset	Time (s)	CDCNN	DBMA	SSRN	DBDA	CEGCN	A^2S^2K-ResNet	SSFTT	FDGC	GTFN	AMGCFN	AMS-M2ESL	MFAF	$MFAF_b$
Indian Pines	Train.	12.81	25.97	21.07	28.02	25.66	18.53	7.68	8.86	174.43	**4.87**	59.65	12.63	10.89
	Test.	2.20	6.12	3.47	5.77	14.52 ms	3.52	0.50	0.92	10.75	12.17 ms	0.28	6.48 ms	**4.75 ms**
Pavia University	Train.	7.04	10.92	11.08	16.25	66.83	9.66	**4.21**	4.80	32.48	164.10	7.69	110.10	75.56
	Test.	10.18	16.68	13.20	17.73	**14.41 ms**	10.97	2.13	4.41	41.61	48.67 ms	40.43 ms	27.30 ms	24.07 ms
Salinas	Train.	9.20	17.26	15.58	24.98	32.31	19.23	5.07	**4.90**	35.81	85.01	22.25	54.30	41.57
	Test.	11.78	33.69	17.59	32.07	**6.74 ms**	13.36	2.70	6.02	54.42	28.13 ms	61.03 ms	15.57 ms	13.12 ms
Houston 2013	Train.	4.00	8.04	5.69	9.68	OOM	5.84	**2.99**	4.93	31.90	OOM	4.93	OOM	124.50
	Test.	3.44	7.08	4.97	7.02	OOM	4.06	0.83	1.57	15.34	OOM	0.31	OOM	**34.78 ms**
	Train. (CPU)	**5.59**	77.47	56.84	148.50	507.46	10.67	8.68	10.09	44.55	2557.88	5.75	3179.03	2151.13
	Test. (CPU)	7.61	179.73	103.79	222.43	1.93	13.00	4.64	7.51	33.83	8.95	**0.45**	7.95	4.84

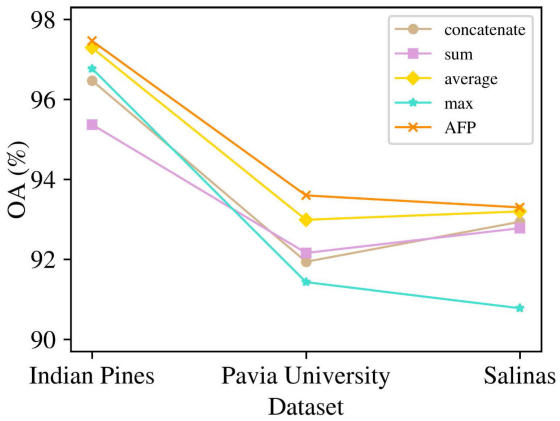


Fig. 9. Results of different fusion schemes on Indian Pines, Pavia University, and Salinas datasets.

is due to the noise suppression network designed by the three models. In addition, our MFAF outperforms the comparison methods, which effectively suppresses the noise and improves the performance.

E. Analysis of the Proposed AFP and the Compared Fusion Schemes

For simplicity, we utilize concatenate, sum, average, and max to represent column concatenation fusion, sum fusion, average fusion, and maximum fusion, respectively. Fig. 9 shows the OA results generated by different fusion schemes on Indian Pines, Pavia University, and Salinas datasets. From the figure, we observe that the proposed AFP achieves the highest OA over all compared fusion schemes. This demonstrates the effectiveness of designing AFP.

F. Comparison of Running Time

In this section, we compare the training and testing time of different methods on all datasets, and the results are listed in Table IX. From the results, we make the following observations First, Compared with these deep networks, i.e., DBMA, SSRN, DBDA, and A^2S^2K-ResNet, CDCNN with several 1×1 convolutional layers is faster in terms of training and testing time. Second, SSFTT achieves the SOTA training time on Pavia University and Houston 2013 datasets, which can quickly learn on the two datasets. Third, on large dataset (Houston 2013), AMS-M2ESL performs faster than most methods in terms of training and testing time. These results demonstrate the efficiency of AMS-M2ESL on large HSI datasets. Fourth, CEGCN, AMGCFN, and the proposed MFAF and $MFAF_b$ use the whole HSI as input and perform the computation in parallel. Therefore, their prediction speed is relatively faster than other methods that take small HSI cubes as input. However, on the one hand, the four methods require a significant amount of training time to read and write a large number of pixels with whole HSI. On the other hand, the first three methods require substantial memory resources on Houston 2013 with large dataset, which leads to the memory overflow on GPU when calculating the adjacency and association matrices of superpixel graph. Fifth, MFAF takes slightly longer than CEGCN on most datasets because of the design of dual branch structure in the proposed CNN, and performs better in terms of OA. Sixth, on the large dataset, compared with baselines (not include our MFAF), $MFAF_b$ achieves the best performance in terms of inference speed (GPU device) and classification accuracy. These observations show the superiority of our methods for applications in the industrial world.

G. Ablation Study

In this section, we validate the influences and contributions of these modules under different numbers of training samples through experiments: different networks, node's own features, attention mechanism, dual branch, and skip connections (residual connections).

1) Contributions of Different Networks: As introduced in Fig. 1, the proposed MFAF is mainly composed of three networks, i.e., the spectral interaction network, the multiscale attentional GCN, and complementary dual convolutional attention network. To evaluate the contributions of these networks, we compare the OA results generated by the full MFAF with those generated by the versions of MFAF that omit one of the three networks. For clarity, we use $MFAF_a$, $MFAF_b$, and $MFAF_c$ to denote the modified methods of MFAF that

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TABLE X

OA COMPARISON OF MFAF AND MFAF WITHOUT ONE OF THE THREE NETWORKS UNDER VARIOUS PERCENTAGES OF TRAINING SAMPLES PER CLASS ON INDIAN PINES, PAVIA UNIVERSITY, AND SALINAS DATASETS

Method	Indian Pines			Pavia University			Salinas		
	1%	2.5%	5%	0.05%	0.1%	0.5%	0.05%	0.1%	0.5%
$MFAF_a$	86.23	95.81	98.21	70.86	91.12	98.38	78.08	88.46	97.34
$MFAF_b$	83.50	95.57	98.47	72.78	87.35	98.82	77.16	86.80	97.71
$MFAF_c$	76.43	81.99	83.07	70.73	86.78	91.65	83.54	89.59	94.70
MFAF	**90.33**	**97.46**	**98.76**	**80.97**	**93.60**	**99.37**	**85.07**	**93.30**	**98.95**

TABLE XI

OA COMPARISON OF MFAF AND MFAF WITHOUT S_a^0 UNDER VARIOUS PERCENTAGES OF TRAINING SAMPLES PER CLASS ON INDIAN PINES, PAVIA UNIVERSITY, AND SALINAS DATASETS

Method	Indian Pines			Pavia University			Salinas		
	1%	2.5%	5%	0.05%	0.1%	0.5%	0.05%	0.1%	0.5%
MFAF without S_a^0	90.16	97.28	98.71	80.73	93.08	99.34	84.74	93.29	98.92
MFAF	**90.33**	**97.46**	**98.76**	**80.97**	**93.60**	**99.37**	**85.07**	**93.30**	**98.95**

TABLE XII

OA COMPARISON OF MFAF AND THE REDUCED MFAF METHODS THAT REMOVE AT LEAST ONE ATTENTION MECHANISM UNDER VARIOUS PERCENTAGES OF TRAINING SAMPLES PER CLASS ON INDIAN PINES, PAVIA UNIVERSITY, AND SALINAS DATASETS

Method	Indian Pines			Pavia University			Salinas		
	1%	2.5%	5%	0.05%	0.1%	0.5%	0.05%	0.1%	0.5%
$MFAF_d$	89.43	97.06	98.69	79.32	92.30	99.25	83.87	91.99	98.69
$MFAF_e$	82.28	94.60	98.05	77.31	91.49	99.30	84.59	92.03	98.61
$MFAF_f$	80.98	94.01	98.03	75.73	90.69	99.23	83.51	91.51	98.56
MFAF	**90.33**	**97.46**	**98.76**	**80.97**	**93.60**	**99.37**	**85.07**	**93.30**	**98.95**

exclude the spectral interaction network, the multiscale attentional GCN, and complementary dual convolutional attention network, respectively. The results are reported in Table X. We observe that MFAF achieves higher OA under various numbers of training samples over all modified methods. In addition, MFAF outperforms the modified methods of MFAF by a substantial margin under small training samples. These results validate the contributions of the three networks to performance improvement.

2) Contributions of Node's Own Features: As shown in (15), we introduce additional node's own features via S_a^0. To verify the impact of node's own features, we compare the OA results produced by MFAF with those of MFAF excluding S_a^0 under various numbers of training samples, and the results are listed in Table XI. From the results, as the number of training samples decreases, MFAF increasingly outperforms MFAF without S_a^0. This indicates that self-nodes are more important than neighboring nodes to improve performance under small training samples.

3) Contributions of Attention Mechanism: In the proposed MFAF, we design the two attention mechanisms for the multiscale attentional GCN and complementary dual convolutional attention network. To evaluate the significance of these attention mechanisms, we compare the OA results of MFAF with those of modified MFAF version that exclude at least one of the attention mechanisms. For simplicity, we utilize $MFAF_d$, $MFAF_e$, and $MFAF_f$ to denote the modified versions of MFAF that exclude the attention mechanism of the proposed multiscale GCN, the attention mechanism of the proposed dual branch CNN, and all attention mechanisms, respectively. Table XII reports the results. We see that $MFAF_d$ and $MFAF_e$ consistently outperform $MFAF_f$, and are lower than MFAF under small training samples. This demonstrates that the two attention mechanisms with small training samples contribute to the performance improvement.

4) Contributions of Dual Branch: In the proposed dual branch CNN, we design the dual branch structure (branch 1 and branch 2) to improve the classification performance. To verify the contributions of these two branches, we compare the OA results of MFAF with those of MFAF versions that exclude one of the two branches. $MFAF_g$ and $MFAF_h$ represent the modified methods that exclude branch 1 (which includes branch 1_1 and branch 1_2) and branch 2 (which includes branch 2_1 and branch 2_2), respectively. Table XIII lists the results. From the table, we observe that MFAF achieves the best OA over $MFAF_g$ and $MFAF_h$ on all datasets and is significantly higher than the modified methods under small training samples on most datasets. The results support the design of dual branch structure.

5) Contributions of Skip Connections: In the proposed MFAF, we develop two pairs of symmetrical skip connections in the proposed attention-based residual block. To verify the impacts of the two pairs of skip connections, we compare the OA results generated by MFAF with the MFAF versions that

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TABLE XIII

OA COMPARISON OF MFAF AND MFAF WITHOUT ONE OF THE TWO BRANCHES UNDER VARIOUS PERCENTAGES OF TRAINING SAMPLES PER CLASS ON INDIAN PINES, PAVIA UNIVERSITY, AND SALINAS DATASETS

Method	Indian Pines			Pavia University			Salinas		
	1%	2.5%	5%	0.05%	0.1%	0.5%	0.05%	0.1%	0.5%
$MFAF_g$	84.15	92.65	96.18	77.31	91.82	99.09	82.68	91.50	98.60
$MFAF_h$	90.33	97.41	98.63	80.10	92.96	99.32	84.55	92.38	98.84
MFAF	**90.33**	**97.46**	**98.76**	**80.97**	**93.60**	**99.37**	**85.07**	**93.30**	**98.95**

TABLE XIV

OA COMPARISON OF MFAF AND THE REDUCED MFAF METHODS THAT REMOVE AT LEAST ONE PAIR OF SKIP CONNECTIONS UNDER VARIOUS PERCENTAGES OF TRAINING SAMPLES PER CLASS ON INDIAN PINES, PAVIA UNIVERSITY, AND SALINAS DATASETS

Method	Indian Pines			Pavia University			Salinas		
	1%	2.5%	5%	0.05%	0.1%	0.5%	0.05%	0.1%	0.5%
$MFAF_i$	88.98	97.24	98.61	80.45	93.18	99.37	84.50	92.94	98.86
$MFAF_j$	89.96	97.07	98.30	80.65	93.00	99.10	84.89	92.21	98.86
$MFAF_k$	89.56	97.15	98.16	80.10	92.33	99.19	84.75	93.24	98.76
MFAF	**90.33**	**97.46**	**98.76**	**80.97**	**93.60**	**99.37**	**85.07**	**93.30**	**98.95**

omit at least one pair of skip connections. We use $MFAF_i$, $MFAF_j$, and $MFAF_k$ to denote the modified versions of MFAF that exclude the first pair of skip connections, the second pair of skip connections, and all skip connections, respectively. The results are summarized in Table XIV. With small training samples, MFAF outperforms the modified MFAF methods, which shows the advantages of the proposed attention-based residual block constructed by the two pairs of skip connections under small training samples.

V. CONCLUSION

In this work, we introduce a novel HSI classification framework combining multiscale attentional GCN and dual-branch attention CNN to learn the complementary spectral–spatial features. In the GCN, we explore a multiscale attentional adjacency matrix to model the long-range correlations between land covers. For the CNN, with these DL technologies of attention mechanism, dual branch, residual connection, DSC, and dilated convolution, we propose a novel complementary dual convolutional attention network to learn the local spectral–spatial features of pixels. Furthermore, the AFP fusion is used to fuse the complementary features captured by the GCN and CNN, namely spectral–spatial features and pixel- and superpixel-level features. Extensive experiments conducted on four HSI datasets prove the effectiveness and advantages of the proposed MFAF. For future work, an interesting direction is to explore the selection and modeling of spectral bands.

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Xun Liu received the M.Sc. degree from the School of Electronics and Information, Guangdong Polytechnic Normal University, Guangzhou, China, in 2020. He is currently pursuing the Ph.D. degree with the School of Information Engineering, Guangdong University of Technology, Guangzhou, China.

He has published more than ten papers in refereed journals and conference proceedings. His research interests include hyperspectral image classification, deep learning, and graph deep learning.



Alex Hay-Man Ng (Senior Member, IEEE) received the B.E. and M.Eng.Sc. degrees in electrical engineering and the Ph.D. degree in surveying and spatial information systems from the University of New South Wales (UNSW), Sydney, NSW, Australia, in 2004, 2005, and 2011, respectively.

From 2010 to 2016, he was a Research Associate at UNSW. He is currently a Full Professor with the School of Civil and Transportation Engineering, Guangdong University of Technology, Guangzhou, China. His research encompasses a wide range of areas within geosciences, such as satellite remote sensing, InSAR algorithm and software, ground subsidence monitoring and modeling, and image classification.



Linlin Ge (Member, IEEE) received the B.E. degree (Hons.) in optical engineering from Wuhan Technical University of Surveying and Mapping, Wuhan, China, in 1985, the M.S. degree in crustal deformation from the Institute of Seismology, China Earthquake Administration, Wuhan, in 1988, and the Ph.D. degree in GPS and remote sensing from the University of New South Wales (UNSW), Sydney, NSW, Australia, in 2001.

From 1997 to 1998, he was a Postdoctoral Research Fellow at the Meteorological Research Institute, Japan Meteorological Agency sponsored by the Science and Technology Agency of Japan. He is currently a Professor of remote sensing and Earth observation at the School of Civil and Environmental Engineering, UNSW. He has been studying Earth observation techniques since 1985.

Prof. Ge is a Foundation Member of the Surveying and Spatial Sciences Institute. He is a Life Member of the American Geophysical Union (AGU). Among many other prestigious awards, he and his team's work to support the Sichuan Earthquake rescue effort won them the highly prestigious JK Barrie Award for Overall Excellence at the 2008 Asia–Pacific Spatial Excellence Awards. He was named "NSW Scientist of the Year 2009" in the category Physics, Earth Sciences, Chemistry and Astronomy for his work in near real-time satellite remote sensing. He is also a Principal Investigator for satellite missions such as ENVISAT, ALOS, COSMO-SkyMed, Radarsat-2, and ALOS-2.



Fangyuan Lei (Member, IEEE) received the B.S. and Ph.D. degrees from Northwestern Polytechnic University, Xi'an, China, in 1998 and 2004, respectively.

He is currently a Professor with the School of Electronic and Information, Guangdong Polytechnic Normal University, Guangzhou, China. He is also a member of Guangdong Provincial Key Laboratory of Intellectual Property Big Data. His research interests include graph convolutional network, image processing, and intellectual property big data.



Xuejiao Liao is currently pursuing the M.Sc. degree with Guangdong Provincial Key Laboratory of Intellectual Property Big Data, Guangdong Polytechnic Normal University, Guangzhou, China.

Her research interests include hyperspectral image classification and deep learning.

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